The Buffering Code

Genome-duplication history reorganizes the immune-disease genetics of paralogous transcription factors

iteration15 — human-only paralogous-TF duplication analysis

2026-07-01

1,153 paralogous TFs $\times$ {provenance, arrangement} $\times$ {PBMC, spleen, cytokine, dosage, monogenic, polygenic}

The one idea

The unit of immune-disease genetics may not be the *gene* but the **buffered paralog pair**.

Two independent axes of a gene's duplication history:

- **PROVENANCE** — **2R whole-genome-duplication ohnolog** vs. small-scale duplicate (SSD)
- **ARRANGEMENT** — **clustered** (a co-regulated neighbor buffers the sibling) vs. **dispersed**

	clustered (buffered)	dispersed (unbuffered)
ohnolog	**clustered_cis_ohnolog** (106)	**dispersed_ohnolog** (524)
SSD	**clustered_SSD_tandem** (73)	**dispersed_SSD** (450)

KRAB-ZNF + HOX arrays removed up front. `dup_class4` in `TF_dup_2x2_classification.tsv`.

Design: one denominator, one test

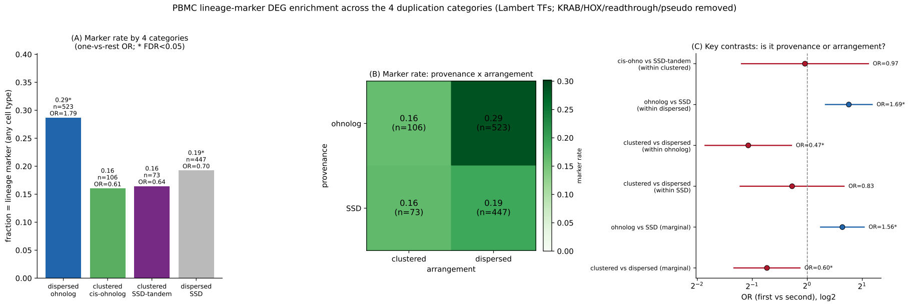
- **Universe:** the 1,153 classified paralogous TFs (or its detectable subset $\cap$ single-cell `var_names`).
- **Test:** for each category, a one-vs-rest Fisher exact odds ratio (OR, 95% CI, BH-FDR) *within* that universe.
- **Contrasts:** **ohnolog vs SSD** (provenance) and **clustered vs dispersed** (arrangement/buffering).

Six readouts, one framework

PBMC identity · spleen identity · cytokine response · dosage buffering · **monogenic disease (IEI)** · **polygenic disease (GWAS)** · **dosage sensitivity (MRDS)**

The recurring question: *is a signal a property of PROVENANCE (being an ohnolog) or of BUFFERING (being clustered)?*

Cell-identity DEGs: a property of *dispersed* ohnologs



PBMC identity markers

- **dispersed_ohnolog**: 28.7%, **OR=1.79** (FDR-sig)
- clustering *erases* it: within-ohnolog clustered-vs-dispersed **OR=0.47**

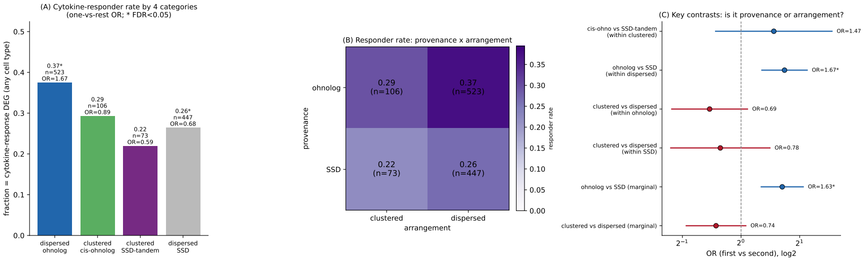
Spleen identity markers (reproduces)

- **dispersed_ohnolog**: **OR=2.87** (p=2e-7)
- within-ohnolog clustered-vs-dispersed **OR=0.33**

Identity is carried by **unbuffered** ohnologs; a buffering neighbor cancels the signal.

Cytokine response: ohnolog provenance *survives* clustering

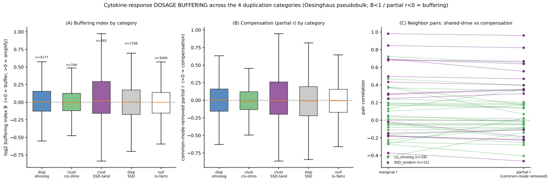
PBMC CYTOKINE-RESPONSE DEG enrichment across the 4 duplication categories (Oesinghaus dictionary; Lambert TFs; KRAB/HOX/readthrough/pseudo removed)



- **dispersed_ohnolog** most enriched (37.5%, **OR=1.67**, FDR-sig).
- BUT **clustered_cis_ohnolog** stays high (29.2%, ns vs dispersed) — unlike identity.
- marginal ohnolog-vs-SSD **OR=1.63** (p=2e-4); clustered-vs-dispersed **ns**.

Clustering converts a 2R ohnolog from an identity factor into a dedicated cytokine EFFECTOR (the ETS/STAT clustered cis-ohnologs).

Where buffering actually lives: specific clustered pairs



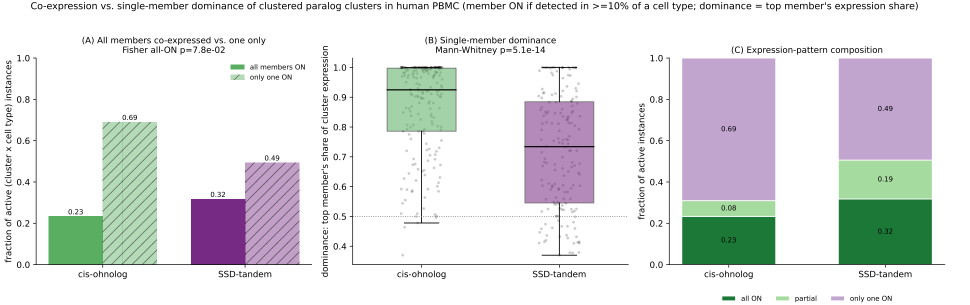
Buffering is *not* a category-level trait (every category median $B \approx 1.0$), but the two **clustered** categories diverge among neighbors:

- **SSD-tandem** neighbors co-**AMPLIFY**: SAND *SP100/110/140/140L* B 1.4–1.8, partial r up to 0.82.
- **cis-ohnolog** harbor the cleanest **BUFFER**: *STAT1/STAT4* B=0.78, partial $r = -0.41$.

cis-vs-SSD neighbor B contrast p≈0.038.

Buffering (STAT) vs amplification (SAND) are the signatures of the two clustered categories.

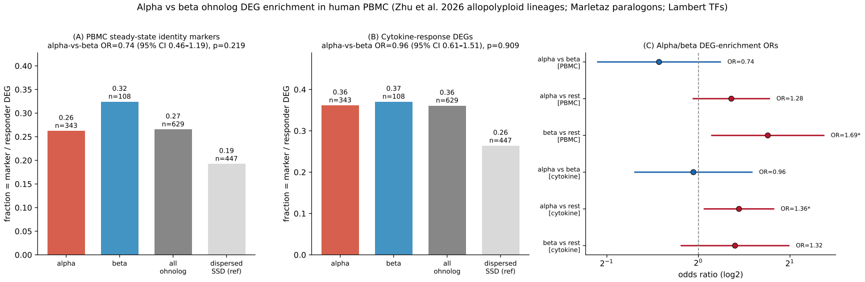
Clustered cis-ohnologs really do co-express (single-cell PBMC)



- **cis-ohnolog** clusters: tighter joint on/off + dominant-member skew (median top-share 0.92 vs 0.73 for **SSD-tandem**; Mann-Whitney p=5e-14).
- *ETS1/FLI1* co-active in all 15 cell types (all-on 1.00, never single) — the archetypal buffered pair.

The empirical readout behind "buffering": a co-regulated neighbor is actually on when its sibling is.

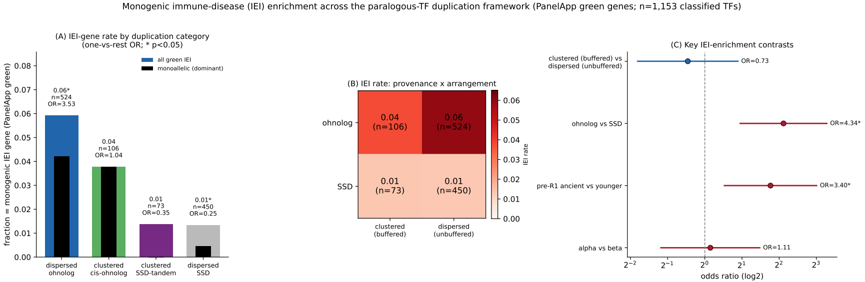
Alpha vs beta ohnologs (Zhu *et al.* Nature 2026)



- 451 detectable ohnolog TFs labeled = 343 alpha / 108 beta (3.2:1 — reproduces the alpha-retention bias).
- Zhu's **ohnolog** > **SSD** REPLICATES in PBMC (both lineages beat dispersed_SSD).
- BUT the **alpha** > **beta** brain asymmetry does NOT: alpha-vs-beta ns for both modalities.

Robust claim = ohnolog>SSD; within-ohnolog alpha/beta contrasts are trends (beta *n* small).

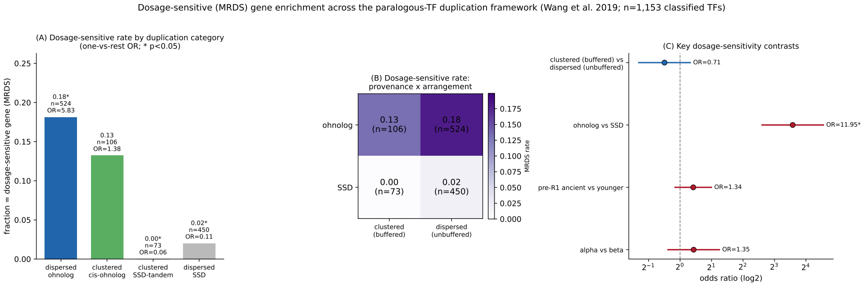
Monogenic immune disease (IEI): a PROVENANCE property



- 42/1,153 paralogous TFs are green IEI genes. **dispersed_ohnolog** **OR=3.53** (p=2e-4).
- **ohnolog**-vs-SSD **OR=4.34** (p=1e-4); pre-R1-ancient-vs-younger **OR=3.40**.
- clustered-vs-dispersed (buffered vs unbuffered) **ns** (OR=0.73).

SPI1/PU.1 + *ELF4* + *GATA2* + *IRF8* are dispersed-ohnolog green IEI genes.

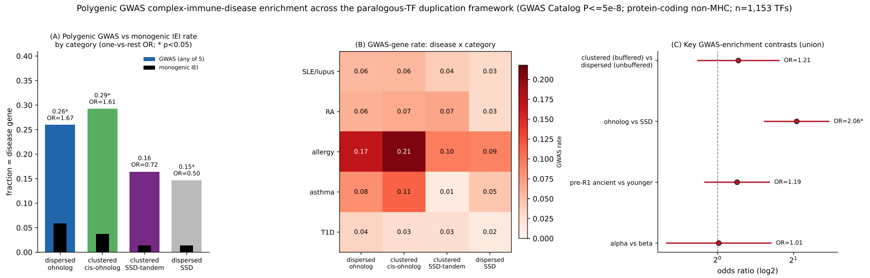
Dosage sensitivity (MRDS): the same PROVENANCE signal, sharper



- 118/1,153 (10.2%) are dosage-sensitive (Wang *et al.* 2019). **dispersed_ohnolog**: 18.1%, **OR=5.83** (FDR=9e-16).
- **ohnolog**-vs-SSD **OR=11.95** (p=6e-21) — the dominant axis. Both SSD groups **depleted** (**tandem** 0/73).
- clustered-vs-dispersed **ns** (OR=0.71); alpha/beta & origin-age ns.

Dosage sensitivity tracks being an ohnolog — not being buffered. (An orthogonal, non-disease annotation.)

Polygenic disease (GWAS): the monogenic → polygenic FLIP



- 245/1,153 (21%) are GWAS genes (5 complex immune diseases).
- **clustered_cis_ohnolog** is the **MOST GWAS-enriched** category (29.2%, **OR=1.61**, p=0.045)...
- ...despite being at **background for monogenic IEI** (3.8%). *ETS1/FLI1/ERG*, *STAT5A*, *BATF* — lupus/allergy loci, not Mendelian.

Buffering converts a monogenic dosage gene into a polygenic risk locus.

The synthesis: two knobs, two disease modes

Readout	disp. ohnolog	clust. cis-ohno	clust. SSD-tan	disp. SSD
PBMC identity (OR)	**1.79***	0.61	0.64	0.70
Spleen identity (OR)	**2.87***	0.51	0.60	0.42
Cytokine response (OR)	**1.67***	0.89	0.59	0.68
Monogenic IEI (OR)	**3.53***	1.04	0.35	0.25
Dosage-sensitive (OR)	**5.83***	1.38	0.06	0.11
Polygenic GWAS (OR)	1.54	**1.61***	0.64	0.49

- **PROVENANCE (ohnolog)** sets **dosage sensitivity, monogenic disease, and cell identity**.
- **BUFFERING (clustering)** *cancels* identity/monogenic signal and **shifts disease to polygenic**.
- **SSD** is dosage-*insensitive* throughout; tandem SSD amplifies rather than buffers.

* FDR / $p < 0.05$, one-vs-rest within the paralogous-TF universe.

Why it matters

- A single evolutionary axis — **2R ohnolog provenance** — predicts which paralogous TFs are **dosage-sensitive** and **monogenic-disease** genes, *before any patient is sequenced*.
- A second axis — **cis-clustering / buffering** — **re-routes** the same class of gene from Mendelian to polygenic (the *ETS1*/*FLI1* autoimmunity pole).
- Suggests buffering is a measurable, cell-type-specific variable behind penetrance and the immunodeficiency $\leftrightarrow$ autoinflammation $\leftrightarrow$ autoimmunity spectrum.

Caveats: enrichment is WITHIN the paralogous-TF universe (not genome-wide), denominators are not expression-matched; provenance uses conservative OHNOLOGS v2 + cis-rescue; GWAS is nearest-gene, MHC-excluded; small clustered $n \Rightarrow$ within-clustered contrasts are trends.

The buffering code

"which buffered network failed, in which cell, and how do we re-balance it?"

iteration15 · scripts 01–17 · `TF_dup_2x2_classification.tsv` + 8 enrichment analyses